

Notes Installation Guide
Version V 1.0

What is Notes

Notes is a web-based browser that works with the Musica database to provide an easy-to-use, clean, straightforward UI/UX experience. It is the human-facing tool to the sqlite database back-end. It has two modes that are very similar in operation:

- Personal – used in conjunction with a personal Musica install, either locally or on a mounted network drive. The personal version of Notes uses a python web-server to act as the communications pipeline between the back-end db and the user.
- Network/Secure – Intended for use on a web-server as either a personal database or a commercial store front that is accessible via a web page. This version of Notes was developed using Apache/2.4.67. In turn, it also has a simple security protocol that will be explained in greater detail in the Musica/Notes User Guide

Both versions can access the same musica.db, depending on how the Notes config files are written.

Installation instructions

To install Notes, follow the steps below:

Download the Musica from the GitHub project database:

(<https://github.com/jcipale/Notes/RELEASE>). Select the ‘Download’ icon from the github interface. The Notes_V1.0 tar-file will be downloaded into your Downloads directory.

Create the installation directory:

Installing Notes is dependent on the following:

- Musica has been installed and setup (Personal install).
- The appropriate web-server has been configured to serve web-pages locally or publicly.

The following assumptions therefore apply:

- For a personal installation, the Musica directory exists in¹ <HOME>/Musica or <PATH>/Musica.

¹ As explained in the Musica Installation Guide – HOME is the users local path. I.E. /home/<user_id>/Musica (linux/MacOS) or C:\Users\<user_id>\Musica on Windows. PATH is a mounted network drive path. I.E. /mnt/<mount_port>/Musica (linux/MacOS) or a [E:](#)\<drive_share>\Musica on Windows.

- For a domain-based installation, check against the specific OS the web-server you are using. Debian (on which Notes was developed) defines the DOCUMENT_ROOT as /var/www/<domain_name>. Other web-servers (such as IIS or BusyBox) may use a different path for DOCUMENT_ROOT. Consult your web admin or documentation for the correct location.

Before installing Notes, perform the following steps:

- locate the Musica installation (personal) OR the DOCUMENT_ROOT (hereafter known as ROOT) on the web server.
- From the command-line, navigate into the Musica directory and look for the folder named 'web'. Web is the destination directory for Notes. For the domain based version, access the server² where the web-server application is installed. Navigate to the ROOT directory and create a directory titled musica-notes.com³ or whatever domain name you wish to use.

If the Notes tarball has not been downloaded, go to <https://github.com/jcipale/Notes/RELEASE>. Select the 'Download' icon from the github interface. Look for/download the Notes_V1.0 tar-file. Click the 'Download' icon and Notes will be retrieved into the Downloads directory. Once Notes has been downloaded:

- Copy the tar file Notes_v1.0.tar.gz to the web location within Musica:
% cd <HOME>/Musica/web
or
% cd <PATH>/Musica/web
% cp <HOME>/Downloads/Notes_V1.0.tar.gz .

² The user must have access to the server or have the systems administrator setup the Notes software under the assigned ROOT.

³ Note that the index page refers to Musica-Note, but the domain name itself can be any domain the user chooses.

- From a terminal window, issue the commands -
% gunzip Notes_v1.0.tar.gz
% tar xvf Notes_v1.0.tar.

The tarfile will expand as shown:

```
.
├── config
│   └── config-setup.dta
├── etc
│   └── security_master.txt
├── index.html
├── install
│   └── musica_db_network_install.py
├── Notes_v1.0.tar
├── py
│   ├── authenticate.py
│   ├── server.py
│   └── settings.py
├── README.txt
└── VERSION
```

Once the tar file has been unpacked, the Notes_V1.0.tar file can be deleted or move to an archive location.

- For installing on a web server, copy the Notes tar-ball to the ROOT directory for the intended domain.
% cp <DOWNLOAD_PATH>/Notes_V1.0.tar.gz /<ROOT>/<Domain_name>/.
% gunzip Notes_v1.0.tar.gz
% tar xvf Notes_v1.0.tar.

The unpacked contents are shown above.

After the tar ball is unpacked, the application needs to be configured. To do this, edit the config/config-setup.dta file and the etc/security_master.txt file (if the Notes application is being used on a domain server for public access). These files are used by Notes to access/communicate with/modify the musica database. The security_master.txt file sets the password for the Commercial application.

Edit the config-setup.dta by setting the following values:

- #MASTER_IP=127.0.0.1 – If the intended usage is personal, leave the MASTER_IP set to 127.0.0.1. This IP is used by the python-server internally. Remove the leading ‘#’ to set the IP.

- #DPATH=/home/<user_id>/Musica – Use this setting if the db is local to the user’s home folder. Change <user_id> to the users user_id. For instance, if Aaron has a user_id of aaronb, then the path would be DPATH=/home/aaronb/Musica.

#DPATH=/mnt/<data_path>/Musica – Use this path if the db is located on a mounted drive on the network that the user has access to. For example, if the mounted drive is labeled ‘hdd_1’, the path would become DPATH=/mnt/hdd_1/Musica

#DBASE=musica.db – This value would remain the same

Once the path values have been entered, remove the leading “#” from the modified line.

- # MASTER_USER=<user_id-1>
MASTER_USER=<user_id-2>

MASTER_USER is only used if the USAGE_TYPE set to ‘Commercial’. This is the typical setting if this is a public facing page. Anyone can query a database. But only authorized users can access the database (Add/Edit/Delete a record). If Notes is being used as a Personal application, the MASTER_USER is left commented out.

- #USAGE_TYPE=Personal
USAGE_TYPE works in conjunction with MASTER_USER. If the USAGE_TYPE is set to Personal, then Notes is invoked using the python back-end-server to access the musica database. If the usage is Commercial, then it is suggested that the Notes UI is installed/running on a web-server on a host.

After the edits have been made to the users satisfaction, save the file as config.dta.

Once the config.dta file is created/saved and Commercial is the operating model, the user needs to edit the security_master.txt file. From the <ROOT> directory on the domain server, cd to the etc directory:

- % cd <ROOT>/etc
% vi security_master.txt

The security_master.txt file allows the operator ('store' owner, system admin, etc) to create the password⁴ to access the portions of the UI that allow the database to be modified.

After the file has been edited, save it as security.txt. Once the file has been saved, rename the file and modify the file permissions to that ONLY root or Systems administrator can edit/change the file

```
% chmod 600 security.txt
% mv security.txt .security.txt
% cd ..
% chmod 600 etc
% mv etc .etc
```

By renaming the file and directory, this adds a level of obfuscation-security to a domain-based service. Especially if this is being used as means to query a store inventory.

Once the files have been configured and the domain has been published by the system administrator, the web site should be available from any web browser.

⁴ This is a fairly simple security feature. Next generation will utilize a more secure user/password scheme.